

MA385 Part 1: Solving nonlinear equations

1.6: Fixed Point Iteration

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Newton's method can be considered to be a special case of a very general approach called *Fixed Point Iteration* (**FPI**) or *Simple Iteration*.

The basic idea is:

*If we want to solve $f(x) = 0$ in $[a, b]$, find a function $g(x)$ such that, if τ is such that $f(\tau) = 0$, then $g(\tau) = \tau$.
Choose x_0 and set $x_{k+1} = g(x_k)$ for $k = 0, 1, 2, \dots$*

0. Outline

1 Introduction

2 How not to choose g

3 Fixed points and contractions

■ Fixed Point

4 How many iterations?

5 Newton's method as a FPI

6 Exercises

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For more details, see Section 1.4 (Relaxation and Newton's method) of Süli and Mayers, *An Introduction to Numerical Analysis*

Also, Chapter 3 of Epperson:

https://search.library.nuigalway.ie/permalink/f/3b1kce/TN_cdi_askewsholts_vlebooks_9781118730966

1. Introduction

Yet again, we want to solve

Given a function $f(x)$, find $\tau \in [a, b]$ such that $f(\tau) = 0$.

Again, we'll try to find a sequence $\{x_0, x_1, \dots, x_k, \dots\}$, such that $x_k \rightarrow \tau$ as $k \rightarrow \infty$.

In this section, we'll consider *one-step* methods, which, like Newton's method, compute x_{k+1} just from x_k .

The Method is called **Fixed Point Iteration** (FPI):

- ▶ Choose a function g such that, if $f(\tau) = 0$, then τ is a fixed point of g .
- ▶ Choose x_0 , and then iterate with $x_{k+1} = g(x_k)$.

1. Introduction

Example 1.6.1

Suppose $f(x) = e^x - 2x - 1$ and we are trying to find a solution to $f(x) = 0$ in $[1, 2]$. Then we can take $g(x) = \ln(2x + 1)$.

If we take $x_0 = 1$, then we get the following sequence:

k	x_k	$ f(x_k) $	$ \tau - x_k $
0	1.0000	0.2817	2.5643e-01
1	1.0986	0.1972	1.5782e-01
2	1.1623	0.1273	9.4148e-02
3	1.2013	0.0781	5.5092e-02
4	1.2246	0.0464	3.1868e-02
5	1.2381	0.0271	1.8310e-02
6	1.2460	0.0157	1.0479e-02
$\vdots$	$\vdots$	$\vdots$	
10	1.2553	0.0017	1.1079e-03

2. How not to choose g

We have to be quite careful with this method: **not every choice is g is suitable.**

For example, suppose we want the solution to $f(x) = x^2 - 2 = 0$ in $[1, 2]$. We could choose $g(x) = x^2 + x - 2$. Then, if take $x_0 = 1$ we get the sequence:

2. How not to choose g

Before we do that in a formal way, consider the following...

Example 1.6.2

Use the Mean Value Theorem to show that the fixed point method $x_{k+1} = g(x_k)$ converges if $|g'(x)| < 1$ for all x near the fixed point.

2. How not to choose g

This previous example is useful in two ways:

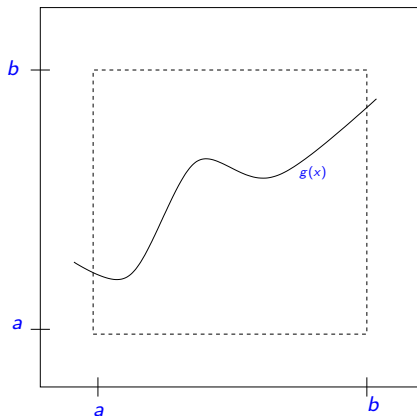
1. It introduces the tricks of using that $g(\tau) = \tau$ & $g(x_k) = x_{k+1}$.
2. It leads us towards the **contraction mapping theorem**.

Definition: fixed point

A function $g : \mathbb{R} \rightarrow \mathbb{R}$ is said to have a **fixed** point at $x = \tau$ if $g(\tau) = \tau$.

Theorem 1.6.1 (Fixed Point Theorem)

Suppose the function g is continuous on $[a, b]$, and $a \leq g(x) \leq b$ for all $x \in [a, b]$. Then $g(x)$ has a *fixed point* in $[a, b]$.



Definition: **Contraction**

Let g be a continuous function defined on $[a, b]$. If there is a number $L \in (0, 1)$ such that

$$|g(\alpha) - g(\beta)| \leq L|\alpha - \beta| \text{ for all } \alpha, \beta \in [a, b], \quad (1)$$

then we say that g is a **contraction** on $[a, b]$.

Theorem 1.6.2 (Contraction Mapping Theorem)

Suppose that the function g is a real-valued, defined, continuous, and

- (a) maps every point in $[a, b]$ to some point in $[a, b]$, and
- (b) is a contraction on $[a, b]$,

then

- (i) $g(x)$ has a fixed point $\tau \in [a, b]$,
- (ii) the fixed point is unique,
- (iii) the sequence $\{x_k\}_{k=0}^{\infty}$ defined by $x_0 \in [a, b]$ and $x_k = g(x_{k-1})$ for $k = 1, 2, \dots$ converges at least linearly to τ .

4. How many iterations?

The algorithm generates as sequence $\{x_0, x_1, \dots, x_k\}$. Eventually we must stop. Suppose we want the solution to be accurate to say 10^{-6} , how many steps are needed? That is, how big do we need to take k so that

$$|x_k - \tau| \leq 10^{-6}?$$

The answer is obtained by first showing that

$$|\tau - x_k| \leq \frac{L^k}{1 - L} |x_1 - x_0|. \quad (2)$$

4. How many iterations?

Example 1.6.3

Suppose we are using FPI to find the fixed point $\tau \in [1, 2]$ of $g(x) = \ln(2x+1)$ with $x_0 = 1$, and we want $|x_k - \tau| \leq 10^{-6}$, then we can use (2) to determine the number of iterations required.

5. Newton's method as a FPI

Newton's method can be thought of as an example of a fixed point method, where we take

$$g(x) = x - \frac{f(x)}{f'(x)}.$$

However, we know that, when Newton's Method converges it does so quadratically, whereas FPI converges (at least linearly).

5. Newton's method as a FPI

Let's remind ourselves of the definition:

- ▶ We have a sequence of numbers $\varepsilon_0, \varepsilon_1, \dots$, such that $\lim_{k \rightarrow \infty} \varepsilon_k = 0$.
- ▶ These bound the errors: $|\tau - x_k| \leq \varepsilon_k$ Let $\tau = \lim_{k \rightarrow \infty} x_k$.
- ▶ We know that $\lim_{k \rightarrow \infty} x_k = \tau$
- ▶ Then we say that the sequence $\{x_k\}_{k=0}^{\infty}$ converges *with at least order* q if

$$\lim_{k \rightarrow \infty} \frac{\varepsilon_{k+1}}{(\varepsilon_k)^q} = \mu,$$

for some constant μ .

For $q = 1$ we get linear convergence. If $q = 2$, we say it is *quadratic*.

5. Newton's method as a FPI

Suppose that we have a convergent Fixed Point Method, $x_{k+1} = g(x_k)$, but with the additional property that $g'(\tau) = 0$. Then, in fact, FPI converges (at least) quadratically):

5. Newton's method as a FPI

Equipped with this, one can show that, in the FPI setting, Newton converges quadratically. See Exercise.

6. Exercises

Exercise 1.6.1

Is it possible for g to be a contraction on $[a, b]$ but not have a fixed point in $[a, b]$? Give an example to support your answer.

Exercise 1.6.2

Show that $g(x) = \ln(2x + 1)$ is a contraction on $[1, 2]$. Give an estimate for L . (Hint: Use the Mean Value Theorem).

6. Exercises

Exercise 1.6.3

Suppose we wish to numerically estimate the famous *golden ratio*, $\tau = (1 + \sqrt{5})/2$, which is the positive solution to $x^2 - x - 1$. We could attempt to do this by applying fixed point iteration to the functions $g_1(x) = x^2 - 1$ or $g_2(x) = 1 + 1/x$ on the region $[3/2, 2]$.

- (i) Show that g_1 is *not* a contraction on $[3/2, 2]$.
- (ii) Show that g_2 is a contraction on $[3/2, 2]$, and give an upper bound for L .

6. Exercises

Exercise 1.6.4

In class we saw that if $g(\tau) = \tau$, and the fixed point method given by

$$x_{k+1} = g(x_k),$$

converges to the point τ (where $g(\tau) = \tau$), and if $g'(\tau) = 0$, then the method converges quadratically.

Show that, in fact if

$$g'(\tau) = g''(\tau) = \cdots = g^{(p-1)}(\tau) = 0,$$

then it converges with order p .

Exercise 1.6.5 (Based on Q2(d), Semester 1 exam, 25/26)

Show that Newton's method can be written as a fixed-point method.

Use the result on Slide 17 to show it converges quadratically.